9.4: Sufficiency

Definition 9.3 : Sufficiency

The statistic $U = g(Y_1, \ldots, Y_n)$ is said to be sufficient for θ if the conditional distribution of $Y_1, \ldots, Y_n$, given U, does not depend on θ.

$\Rightarrow$ That is, if

$$L(\theta \mid U) = \begin{cases} \dfrac{L(\theta)}{p_U(u)}, & \text{for d.r.v.'s} \\ \dfrac{L(\theta)}{f_U(u)}, & \text{for c.r.v.'s} \end{cases}$$

doesn't depend on θ, then U is sufficient for θ.

Example 6:

Let $Y_1, \ldots, Y_n$ denote a r.s. of iid r.v.'s with density function given by

$$f_Y(y) = \begin{cases} \frac{1}{\theta} e^{-y/\theta} & , \ y \geq 0 \\ 0 & , \text{else} \end{cases}$$

j) By conditioning (definition 9.3) is $U = \sum_{i=1}^{n} Y_i$ sufficient for θ?

k) By conditioning (definition 9.3) is $U = Y_{(1)}$ sufficient for θ?

Theorem 9.4 : Factorization

If $L(y_1,...,y_n \mid \theta)$ can be factored into two non-negative functions,

$$\Rightarrow L(y_1,...,y_n \mid \theta) = g(u,\theta) \times h(y_1,...,y_n)$$

where $g(u,\theta)$ is a function only of u and θ and $h(y_1,...,y_n)$ is not a function of θ, then $U = g(Y_1,...,Y_n)$ is a sufficient statistic for θ.

Find the Sufficient Statistic(s)

Example 4: $Y_1, \ldots, Y_n \overset{ind.}{\sim} Poisson(\lambda).$

<u>Example 7:</u> $Y_1, \ldots, Y_n \overset{ind.}{\sim} Normal(\mu_Y, \sigma_Y^2).$

<u>Example 5:</u> $X_1, \ldots, X_n \overset{ind.}{\sim} Uniform(0, \theta).$

9.5: Rao-Blackwell Theorem and MVUE

Theorem 9.5 : Rao - Blackwell Theorem

Let $\hat{\theta}$ be any unbiased estimator for θ where $V(\hat{\theta}) < \infty$. If U is a sufficient statistic for θ, define $\hat{\theta}^* = E[\hat{\theta} \mid U]$.

Then for all θ,

$E[\hat{\theta}^*] = \theta$ and $V[\hat{\theta}^*] \leq V[\hat{\theta}]$.

Example 3:

Consider the r.v. $Y \sim B(n, p)$. Note: $Y = \sum_{i=1}^{n} X_i$, where $X_i \overset{iid}{\sim} Bernoulli(p)$.

e) Consider the unbiased estimator $\hat{p}_1 = X_1$ for p, and the sufficient statistic $U = \sum_{i=1}^{n} X_i$. Use the Rao-Blackwell theorem to find a new (improved) estimator, $\hat{p}_2$.

Example 17 : Suppose $Y_1, \ldots, Y_n$ is a random sample of independent and identically distributed $Poisson(\lambda)$ random variables. Find an unbiased estimator for $\theta = P(Y = 0) = e^{-\lambda}$.